

Reading a Squat Assessment

Learning to Identify Compensation Patterns

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Domain: Movement Analysis & Biomechanics

Topic: Identifying compensation patterns in a side-view squat

Expert: Jolie Abadir (movement analysis and biomechanics background)

User: Sofia Hirao, 19, runner/soccer player, wants to learn proper form to prevent injuries



Domain: Movement Analysis & Biomechanics

Topic: Reading a squat assessment — identifying good vs. compensated form

This topic meets the requirements because squat assessments have clear visual markers (knee cave, forward lean, heel rise) that can be taught and quizzed in 10 minutes using side-by-side video comparison.

Expert: Jolie

- Sofia: Has intuitive body awareness from soccer and running but lacks formal movement analysis vocabulary.
- Sam: Recreational experience in athletics but no formal movement analysis background.

Media: What media would you use?

- 1) Side-by-side videos of good vs. compensated squats, paused at key checkpoints.
- 2) Annotated images highlighting compensation patterns (knee cave, forward lean, heel rise).
- 3) Slow-motion video clips with visual overlays marking joint alignment.

Positive TA Feedback: This seems like a viable topic that could work well as a lesson.

Negative TA Feedback: For anything physical, it may be hard to teach virtually — need to find ways to make it work without hands-on correction.

User:

- Sofia Hirao is a 19-year-old sophomore at Barnard College. She is a runner and former soccer player with intuitive body awareness but no formal movement analysis training.
- Sofia is interested in learning movement analysis to prevent injuries and improve her athletic performance. She has noticed her knees caving inward while running but couldn't identify it as a compensation pattern without guidance.



Quiz:

- Watch a squat video and identify whether it shows good or compensated form.
- Label which compensation pattern is present (knee cave, forward lean, heel rise).
- Given two squat videos, identify which person has better form and explain why.

Who will you run your prototype on?

I would run this one another student in class without formal movement analysis experience.

SquatCheck

Learn to spot compensation patterns in a squat

Start Learning

Subgoal: User chooses to learn or jump to an example. Default: Start Learning.

Next >

What You're Looking For

When someone squats, their body sometimes works around weakness, tightness, or poor mobility. These workarounds are called **compensation patterns**.

They show up as visible deviations from ideal form. Once you know what to look for, you can spot them in real time.

In this lesson, you'll learn to identify 3 compensation patterns visible from a side-view squat:

1. Heel Rise

Heels lift off the ground during the squat

2. Excessive Forward Lean

Torso tilts too far forward relative to the shin

3. Lumbar Flexion (Butt Wink)

Lower back rounds at the bottom of the squat

Learn: 3 Things to Watch (Side View)

1

1. Heel Position

Are the heels flat on the ground, or do they lift up?

If heels rise, it usually means limited ankle mobility.

2

2. Torso-Tibia Alignment

Is the torso roughly parallel to the shin?

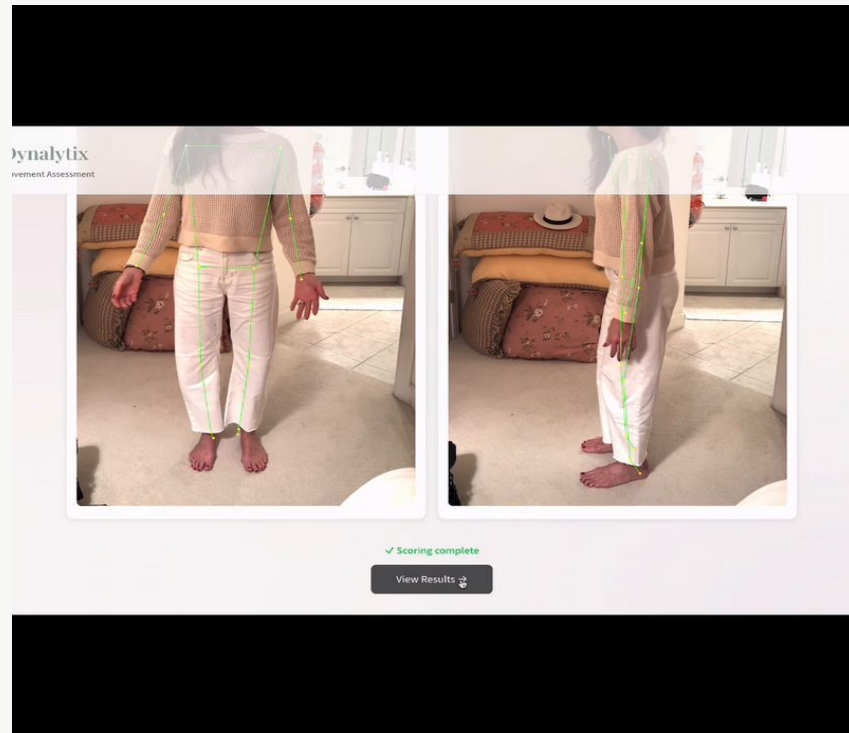
If the torso leans way forward, it suggests tight hips or weak core.

3

3. Lumbar Flexion

Does the lower back stay neutral, or does it round?

Rounding at the bottom (butt wink) indicates limited hip mobility.



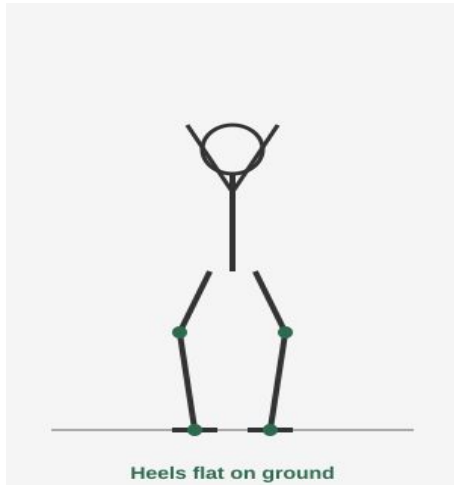
Subgoal: Learn the 3 compensation patterns you'll be identifying.

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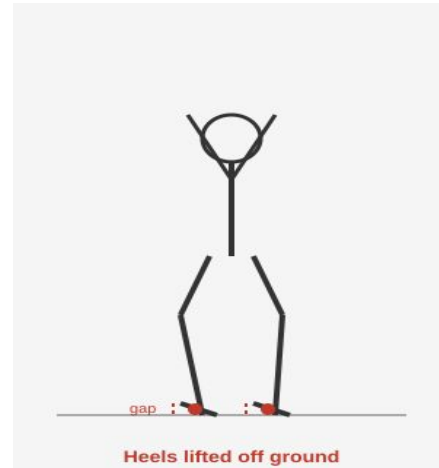
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Learn: Heel Position

GOOD: Heels stay flat



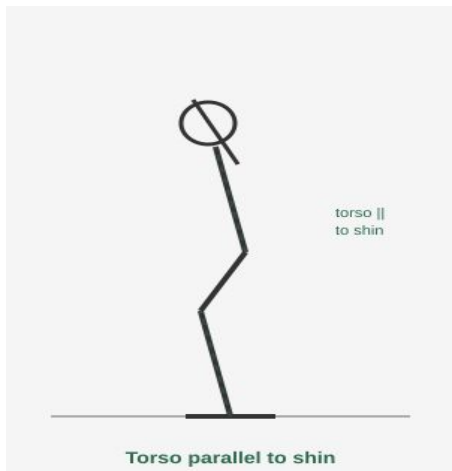
COMPENSATION: Heels lift



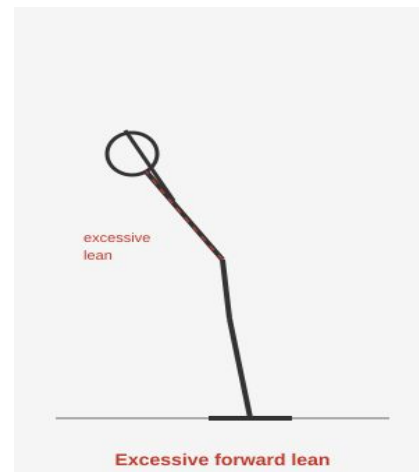
What to look for: Watch the ankles. If heels come up during the squat, that's a compensation for limited ankle mobility.

Learn: Torso-Tibia Alignment

GOOD: Torso parallel to shin



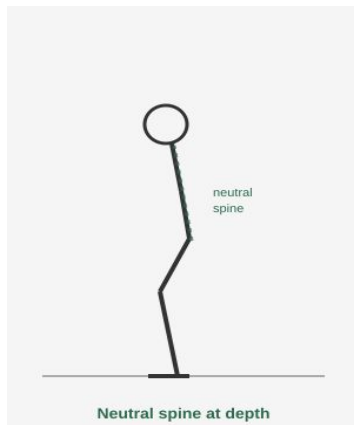
COMPENSATION: Excessive lean



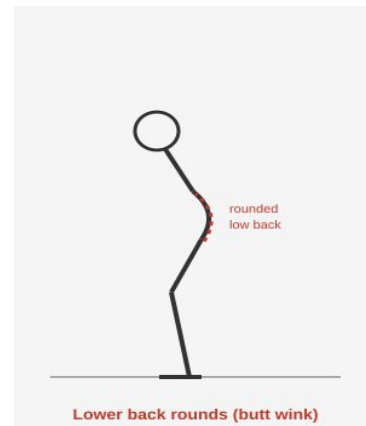
What to look for: Compare the angle of the torso to the shin. In a good squat, they're roughly parallel.

Learn: Lumbar Flexion Control

GOOD: Neutral spine maintained



COMPENSATION: Low back rounds



What to look for: Watch the lower back at the deepest point. If it tucks under and rounds, that's lumbar flexion.

Quick Review

You can now identify 3 compensation patterns visible from a side-view squat:

1. Heel Rise

Heels lift off the ground = limited ankle mobility

2. Excessive Forward Lean

Torso tilts way past the shin = tight hips or weak core

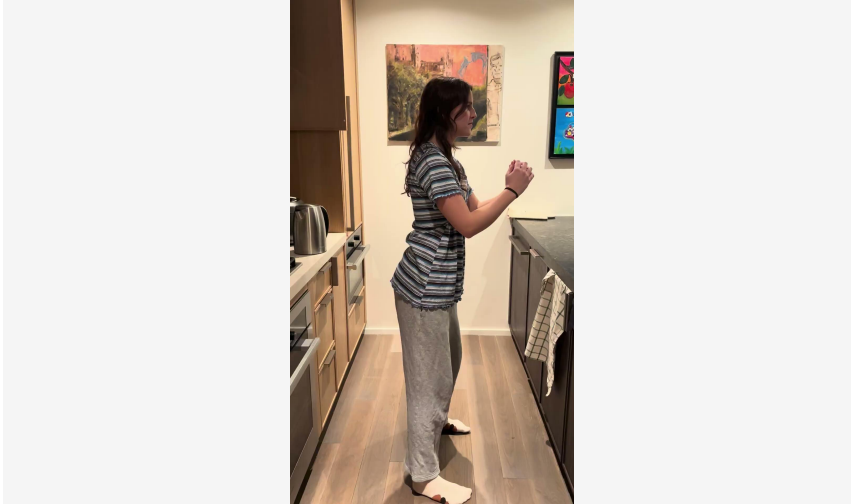
3. Lumbar Flexion (Butt Wink)

Lower back rounds at the bottom = limited hip mobility

Ready to test yourself? Let's go.

Quiz: Question 1 of 3 - Check All That Apply

Watch this side-view squat. Do you see any compensation patterns?



- ☐ Heel Rise
- ☐ Excessive Forward Lean
- ☐ Lumbar Flexion (Butt Wink)
- ☐ No compensations

Submit

Quiz: Answer - Question 1

Correct: Compensations Present

Here's what to look for from the side view:

Heel Position: Heels stay flat. No compensation here.

Torso Alignment: The torso leans much further forward than the shin (30.6 degree difference). Compensation present.

Lumbar Flexion: The lower back rounds at the bottom (45.2 degree trunk angle change). Compensation present.

2 out of 3 side-view compensation patterns are visible.

Quiz: Question 2 of 3

Read this observer's notes. Which compensation pattern are they describing?

Observer notes:

"From the side, I can see that when she reaches the bottom of the squat, her torso is angled way further forward than her shin. Her chest is almost over her toes. Her heels are flat and her lower back looks fine, but her upper body is really far forward."

A) Heel Rise

B) Excessive Forward Lean

C) Lumbar Flexion (Butt Wink)

Subgoal: User applies vocabulary to identify a specific compensation from a written description.

Quiz: Answer - Question 2

Correct: B) Excessive Forward Lean

The key clue: *"her torso is angled way further forward than her shin."*

This describes **torso-tibia alignment** failing. When the torso leans significantly past the shin, that's excessive forward lean.

Why not the others?

- Not heel rise: the observer said "her heels are flat"
- Not lumbar flexion: the observer said "her lower back looks fine"

This often means tight hip flexors or weak core. The body compensates by shifting weight forward to maintain balance.

Quiz: Question 3 of 3

True or False? Mark each statement.

1. *"If someone's heels lift during a squat, it means their knees are weak."*

T

F

2. *"Lumbar flexion means the lower back rounds at the bottom of the squat."*

T

F

3. *"You can check torso-tibia alignment by comparing the angle of the torso to the angle of the shin."*

T

F

4. *"Knee valgus (knees caving in) is best assessed from the side view."*

T

F

Submit

Quiz: Answer - Question 3

1. FALSE *"Heel rise means their knees are weak."*

Heel rise indicates limited ankle mobility (dorsiflexion), not knee weakness. The ankles can't bend enough, so the heels compensate by lifting.

2. TRUE *"Lumbar flexion = lower back rounds at bottom."*

Correct. When the pelvis tucks under and the lower back loses its neutral curve at depth, that's lumbar flexion (butt wink).

3. TRUE *"Torso-tibia alignment = compare torso to shin."*

Correct. In a good squat, the torso and shin should be roughly parallel from the side view.

4. FALSE *"Knee valgus is best assessed from the side."*

Knee valgus (knees caving inward) is only visible from the front view. You can't see medial/lateral knee movement from the side.

User Testing Insights

Tested on: Sofia Hirao (19, runner/soccer player, no formal movement analysis training)

1. **[Positive]** Sofia immediately connected heel rise to her own experience. She said she's seen people at the gym whose heels come up and always wondered why. The stick figure diagrams helped her understand what to look for before seeing real video.
2. **[Negative]** She struggled to distinguish forward lean from lumbar flexion on the quiz. She could tell "something was off" but mixed up the two patterns when asked to name them specifically. The concepts need clearer visual differentiation.
3. **[Positive]** Sofia said the pass/fail format made it easy to compare, she liked being able to look back and forth between good and bad form on the same screen.
4. **[Negative]** On the "check all that apply" quiz question, she hesitated for a long time. She said she wasn't confident enough to commit to an answer without being able to replay the video. Needs a replay button or the ability to scrub through.
5. **[Positive]** She asked if she could try it on her own squat. The idea to "upload your own video" was the most exciting part for her.

Was the user able to make it through? Yes, Sofia completed all 3 quiz questions and the full lesson.

Was the topic too hard, too easy, or about right? About right, she got 2 out of 3 correct, which suggests the material is learnable but not trivial.